



LABORATORY PROJECT REPORT

PLC INTERFACING WITH MICROCONTROLLER AND PC OVER ETHERNET/IP

WEEK 7

SECTION 1

GROUP 7

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ABSTRACT

In this project, an Arduino microcontroller and the OpenPLC Editor were utilized to design, simulate, and implement a Programmable Logic Controller (PLC) control system based on ladder logic programming. The main objective of the experiment was to develop and test a basic industrial control application involving push buttons and an LED through the integration of software and hardware components. The experiment began with the implementation of a simple LED blinking program before progressing to a Start-Stop control circuit incorporating a latching mechanism. Standard PLC elements such as contacts, coils, and timer functions were applied within the ladder diagram to simulate industrial automation logic.

The developed ladder program was compiled and uploaded to the Arduino board after all variables and physical input/output addresses were correctly configured using OpenPLC. The hardware setup was then assembled and tested to verify the responsiveness and reliability of the system under real-time operation. The results demonstrated that the Arduino board could successfully function as a low-cost PLC platform capable of executing IEC 61131-3 compliant ladder logic. Overall, the experiment validated the effectiveness of integrating OpenPLC with microcontroller-based hardware for basic industrial automation and control applications.

CONTENTS

1.0 Introduction

2.0 Materials and Equipment

- 2.1 Electronic Components
- 2.2 Equipment and Tools

3.0 Experimental Setup

4.0 Methodology

5.0 Data Collection

6.0 Data Analysis

7.0 Results

8.0 Discussion

9.0 Conclusion

10.0 Recommendations

11.0 References

1.0 INTRODUCTION

Programmable Logic Controllers (PLCs) are widely used in industrial automation due to their reliability, flexibility, and ease of programming in controlling machinery, production processes, and electromechanical systems. With the advancement of open-source technologies, PLC development can now be integrated with low-cost microcontrollers such as Arduino, enabling students to gain practical exposure to automation and industrial control concepts within a laboratory environment. In this experiment, the OpenPLC Editor was utilized as the primary software platform to develop ladder logic programs, simulate system behavior, and interface with an Arduino board acting as the PLC runtime hardware.

The experiment began with the implementation of a simple LED blinking program before progressing to a more advanced Start-Stop control system involving the design, simulation, and execution of ladder-diagram-based control circuits. To perform these tasks, variables were defined and mapped to the corresponding Arduino physical input and output pins before the developed logic was compiled and uploaded through a USB connection. A physical hardware setup consisting of push buttons, resistors, LEDs, jumper wires, and a breadboard was assembled to validate the implemented control logic under real-time conditions. Through this experiment, students gained hands-on experience in PLC programming, hardware interfacing, ladder logic implementation, and the integration of microcontroller-based systems with industrial automation concepts, including basic Ethernet/IP-oriented communication principles.

2.0 MATERIALS AND EQUIPMENT

2.1 Electronic Components

- Arduino Board
- 2 Push Button Switches
- LED
- Resistors (10 k Ω pull-down resistors and 560 Ω current-limiting resistor)
- Jumper Wires

2.2 Equipment and Tools

- OpenPLC Editor Software
- Breadboard
- USB Cable
- Personal Computer / Laptop

3.0 EXPERIMENTAL SETUP

1. The ladder diagram for the Start-Stop control circuit was created using the OpenPLC Editor.
2. All variables required for the ladder logic, including START, STOP, LATCH, and LED output, were defined and configured.
3. The ladder diagram was compiled and simulated in the OpenPLC environment to verify the correctness of the control logic before hardware implementation.
4. The Arduino board was connected to the computer through a USB connection.
5. The correct COM port number and the corresponding physical pin assignments between the OpenPLC variables and Arduino board were selected and configured.
6. The hardware circuit consisting of push buttons, resistors, LED, jumper wires, and breadboard connections was assembled according to the designed circuit diagram.
7. The ladder logic program was uploaded to the Arduino board for real-time execution.
8. The functionality of the Start-Stop control circuit was tested by observing the LED response when the START and STOP push buttons were pressed.

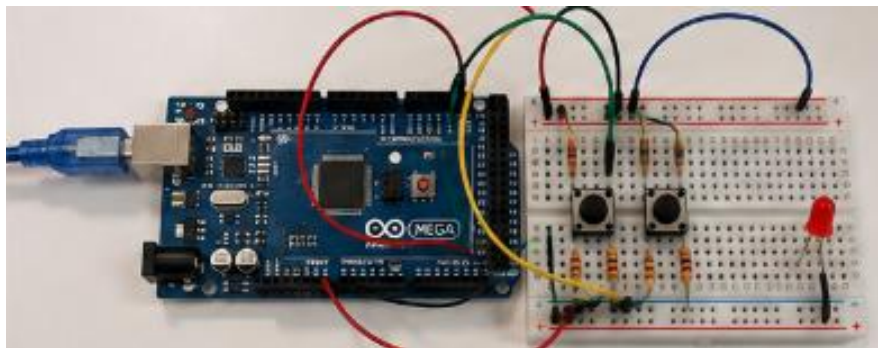
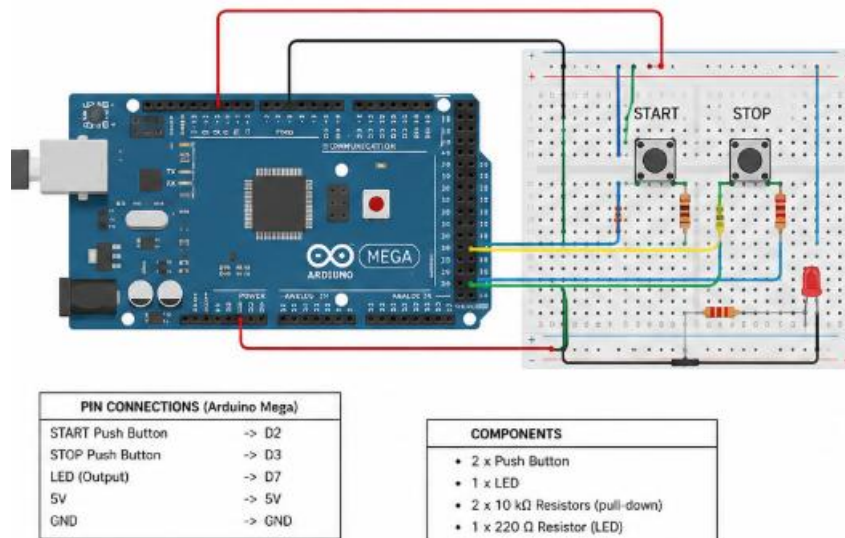


Figure 1 shows the physical circuit

4.0 METHODOLOGY

1. The OpenPLC Editor software was launched, and a new ladder diagram project was created for the Start-Stop control application.
2. Variables for the START push button, STOP push button, LATCH memory bit, and LED output were declared in the variable configuration table with the appropriate BOOL data type.

3. The Start–Stop ladder logic circuit was developed using standard ladder diagram components, including normally open contacts, normally closed contacts, coils, and a latching mechanism.
4. The ladder diagram was compiled and simulated within the OpenPLC Editor to verify the correctness of the control logic and ensure that the LED output responded according to the intended operation.
5. The OpenPLC variables were mapped to the corresponding Arduino physical input and output addresses based on the selected board configuration. The START and STOP push buttons were assigned as digital inputs, while the LED was configured as a digital output.
6. The Arduino board was connected to the computer through a USB cable, and the correct COM port number and board type were selected before transferring the compiled ladder logic program to the microcontroller.
7. The hardware circuit was assembled on a breadboard using push buttons, resistors, jumper wires, and an LED according to the designed circuit configuration. Pull-down resistors were connected to stabilize the input signals from the push buttons.
8. The completed system was tested by pressing the START push button to activate and latch the LED output, while pressing the STOP push button interrupted the latch circuit and turned the LED OFF. The circuit response and system stability were observed throughout the experiment.



Figures 2 show the thinkerCad Circuit connection

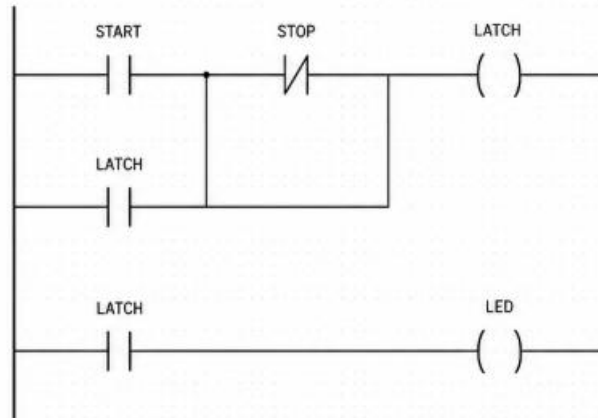


Figure 3 shows the ladder diagram

5.0 DATA COLLECTION

During this experiment, data were collected by observing and evaluating the operation of the Start–Stop control circuit implemented using OpenPLC and an Arduino board. The START and STOP push buttons were tested repeatedly to examine the response of the LED output under different operating conditions. Observations were recorded to determine whether the LED remained ON after the START button was released due to the latching mechanism and whether the STOP button successfully interrupted the circuit and turned the LED OFF immediately.

In addition, the mapping between the OpenPLC variables and the Arduino physical input/output pins was verified to ensure accurate communication between the software and hardware components. The physical circuit connections, including jumper wires, resistors, push buttons, and LED terminals, were also inspected throughout the experiment to prevent unstable signals, loose connections, or irregular circuit behavior. Multiple testing cycles were conducted to evaluate the responsiveness, stability, consistency, and reliability of the control system during real-time operation.

Variable Name	Type	Location (OpenPLC)	Arduino Pin	Function
START	BOOL	%IX0.0	D2	Start push Button (NO)
STOP	BOOL	%IX0.1	D3	Stop push button (NC)
LATCH	BOOL	%M0.0	(Internal)	Internal memory (seal-in)
LED	BOOL	%QX0.0	D7	LED output

Figure 3 shows the function of the PLC

Test	Action	Expected Output	Actual Output
1	Press START	LED ON	Success
2	Release START	LED stays ON	Success
3	Press STOP	LED OFF	Success

Figure 4 shows the data collection

6.0 Data Analysis

Based on the collected data, the performance of the Start–Stop control circuit was analyzed and found to operate according to the intended latching control logic. When the START push button was pressed, the LED output was activated and remained ON even after the button was released. This behavior confirmed that the holding contact implemented within the ladder diagram successfully maintained the output state through the latching mechanism. Conversely, pressing the STOP push button immediately interrupted the latch circuit, causing the LED to switch OFF and confirming the correct execution of the ladder logic program.

The consistent response observed throughout multiple testing cycles indicates that the OpenPLC variable configuration, physical pin mapping, and hardware connections were correctly implemented. In addition, the synchronization between the OpenPLC simulation environment and the Arduino hardware demonstrated reliable communication between the software and physical system. Potential issues such as unstable input signals, loose wiring connections, and input bouncing effects were also considered during testing; however, no major irregularities or timing inconsistencies were detected. Overall, the analysis confirms that the control system operated reliably, accurately, and in accordance with the expected Start–Stop control behavior.

7.0 Results

The experimental results demonstrated that the Start–Stop control circuit functioned successfully under both simulation and hardware implementation conditions. When the START push button was activated, the LED turned ON and remained latched through the holding contact incorporated within the ladder logic program. Pressing the STOP push button consistently deactivated the LED by interrupting the latching circuit.

The OpenPLC simulation produced the same operational behavior as the physical Arduino-based implementation, confirming that the ladder logic, variable configuration, and input/output pin mapping were correctly designed and implemented. Throughout repeated testing cycles, the system maintained stable and consistent performance without noticeable delays, signal instability, or unexpected output behavior. These results indicate that the developed control circuit achieved the objectives of the experiment and successfully demonstrated the practical implementation of a PLC-based Start–Stop control system using OpenPLC and Arduino hardware.

8.0 Discussion

This experiment successfully demonstrated the implementation of a PLC-based Start–Stop control circuit using the OpenPLC Editor and an Arduino microcontroller platform. The system operated according to the expected industrial control logic throughout the testing process. The LED output remained activated after the START push button was released due to the holding contact implemented within the ladder logic, confirming the effectiveness of the latching mechanism commonly used in industrial motor control systems. The STOP push button consistently interrupted the circuit and deactivated the LED output, demonstrating reliable execution of the programmed logic under repeated operating conditions.

The experiment also emphasized the importance of accurate variable assignment and proper physical pin mapping between the OpenPLC environment and the Arduino board. Correct wiring connections and stable input configurations ensured reliable communication between the software and hardware components without significant issues such as unstable outputs, loose wiring, or signal interference. The use of pull-down resistors further improved input signal stability by preventing floating input conditions during operation.

Furthermore, the results demonstrated a strong correlation between the simulation environment and the actual hardware implementation. The OpenPLC simulation accurately represented the real-time behavior of the physical circuit, indicating that the ladder logic program was properly designed and implemented. Overall, this experiment provided valuable practical exposure to ladder logic programming, hardware interfacing, and the integration of microcontroller-based systems within basic industrial automation applications.

9.0 Conclusion

In conclusion, the Start–Stop control circuit was successfully designed, simulated, and implemented using the OpenPLC Editor and an Arduino microcontroller platform. The experimental results confirmed that the control system operated according to the intended ladder logic behavior, where the LED output turned ON when the START push button was pressed and remained latched until the STOP push button interrupted the circuit and switched the LED OFF.

The consistency between the OpenPLC simulation and the physical hardware implementation verified that the ladder diagram, variable configuration, input/output pin mapping, and circuit wiring were correctly implemented. In addition, the experiment provided practical exposure to PLC programming, hardware interfacing, ladder logic development, and real-time control implementation using open-source automation tools. Overall, all objectives of the experiment were successfully achieved, and the experiment enhanced understanding of basic industrial automation and PLC-based control systems.

10.0 Recommendations

For future improvements, several enhancements can be implemented to expand the functionality and practical application of the developed control system. First, additional industrial components such as relays, DC motors, buzzers, or sensors can be integrated into the ladder logic program to simulate more realistic industrial automation processes. This would allow the system to control higher-power devices and perform more advanced automation tasks.

Furthermore, it is recommended to improve the circuit organization and wiring management to reduce the possibility of loose connections, unstable signals, or incorrect pin configurations during operation. The implementation of protective components and external power supplies may also improve the overall stability and reliability of the system when operating under higher electrical loads. Lastly, future experiments may explore more advanced PLC functions such as timers, counters, sequential control, and Ethernet/IP-based communication to further strengthen understanding of industrial automation systems.

11.0 References

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12. Acknowledgments

Special thanks to ZULKIFLI BIN ZAINAL ABIDIN for his sharing the tips and support during this experiment, even though it is a take-home experiment.

Student's Declaration

Certificate of Originality and Authenticity

We hereby certify that we are collectively responsible for the work presented in this report. The content reflects our original work, except where specific references or acknowledgements have been made. No part of this report has been completed or submitted by individuals or sources not identified within.

Furthermore, we confirm that this report is the result of a collaborative effort, with contributions made by all group members. The extent of each individual's contribution is documented within this certificate.

We also hereby certify that we have read and understand the content of the total report and no further improvement on the reports is needed from any of the individual's contributors to the report.

We, therefore, agreed unanimously that this report shall be submitted for marking and this final printed report has been verified by us.

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